

✓ What is Command Prompt?

- Also called **CMD**.
- It's a **text-based interface** in Windows where you can type commands to perform tasks.
- Allows you to manage files, run programs, and configure system settings without using the mouse.
- Commands are mostly simple and used for basic system tasks.

✓ What is PowerShell?

- A more **advanced command-line tool** in Windows.
- Built on top of the .NET framework.
- Supports **complex scripting**, automation, and managing system configurations.
- Can do everything Command Prompt can do **plus much more**.
- Uses **cmdlets** (special commands) and can run scripts.

◆ Main Differences:

Feature	Command Prompt	PowerShell
Interface	Basic command-line	Advanced shell with scripting
Capabilities	Basic file and system commands	Automation, configuration, access to system APIs
Scripting	Limited batch scripting	Powerful scripting language
Target Users	Casual users and beginners	IT professionals and admins

✓ Basic Command Prompt (CMD) Commands

Command	What It Does	Example
dir	Lists files and folders in current directory	dir
cd	Changes directory	cd Documents
mkdir	Creates a new folder	mkdir Projects
rmdir	Deletes an empty folder	rmdir Projects
copy	Copies files	copy file.txt D:\Backup

del	Deletes files	del file.txt
ipconfig	Shows network configuration	ipconfig
ping	Tests network connection to a host	ping google.com
cls	Clears the screen	cls
exit	Closes the CMD window	exit

✓ Basic PowerShell Commands

Command	What It Does	Example
Get-ChildItem (alias ls, dir)	Lists files and folders	Get-ChildItem or ls
Set-Location (alias cd)	Changes directory	Set-Location Documents
New-Item	Creates new file or folder	New-Item -ItemType Directory -Name Projects
Remove-Item	Deletes file or folder	Remove-Item file.txt
Copy-Item	Copies files or folders	Copy-Item file.txt D:\Backup
Clear-Host (alias cls)	Clears the screen	Clear-Host
Test-Connection (alias ping)	Tests network connection	Test-Connection google.com
Get-Process	Lists running processes	Get-Process
Exit	Closes PowerShell	Exit

◆ Note:

- PowerShell commands are **more powerful** and support scripting and automation.
- CMD commands are simpler but still useful for basic tasks.

✓ Windows Troubleshooting Commands

Command	Purpose & Usage
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sfc /scannow	System File Checker: Scans and repairs corrupted system files. Run in Admin mode.
chkdsk	Check Disk: Checks disk for errors and repairs them. Example: chkdsk C: /f /r
ipconfig /release	Releases current IP address from DHCP server. Useful for network reset.
ipconfig /renew	Requests a new IP address from DHCP server.
ipconfig /flushdns	Clears DNS resolver cache. Fixes DNS related issues.
ping [website/IP]	Tests connectivity to a website or IP address. Example: ping google.com
tracert [website]	Traces the route packets take to reach a website. Helps identify network delays.
netstat -an	Shows active network connections and listening ports.
tasklist	Lists all running processes (like Task Manager).
taskkill /PID [id] /F	Forcefully stops a process by its Process ID (PID).
DISM /Online /Cleanup-Image /RestoreHealth	Repairs Windows system image. Run in Admin mode.
systeminfo	Displays detailed system information.
driverquery	Lists all installed device drivers and their status.
mstsc	Opens Remote Desktop Connection tool. Useful for remote troubleshooting.
shutdown /r /t 0	Restarts the computer immediately. Useful after fixing issues.

◆ How to Run These Commands

- Open **Command Prompt** or **PowerShell as Administrator** (right-click → Run as Administrator).
- Type the command and press Enter.

Example for fixing corrupted system files:

```
sfc /scannow
DISM /Online /Cleanup-Image /RestoreHealth
```

✓ What is DISM?

DISM stands for **D**eployment **I**mage **S**ervicing and **M**anagement.

- It's a **Windows command-line tool**.
- Used to **repair and service Windows system images**.
- Helps fix problems when **System File Checker (SFC)** can't repair corrupted system files.

✓ What does DISM do?

- Checks the health of the Windows image (the core system files).
- Repairs missing or corrupted files in the system image.
- Prepares Windows installation files for deployment.

✓ Common DISM commands:

Command	Purpose
DISM /Online /Cleanup-Image /CheckHealth	Checks if the image has corruption (no repair)
DISM /Online /Cleanup-Image /ScanHealth	Scans the image for corruption
DISM /Online /Cleanup-Image /RestoreHealth	Repairs the image if corruption is found

✓ How to use:

1. Open **Command Prompt as Administrator**.
2. Run this command to fix system image issues:

DISM /Online /Cleanup-Image /RestoreHealth